

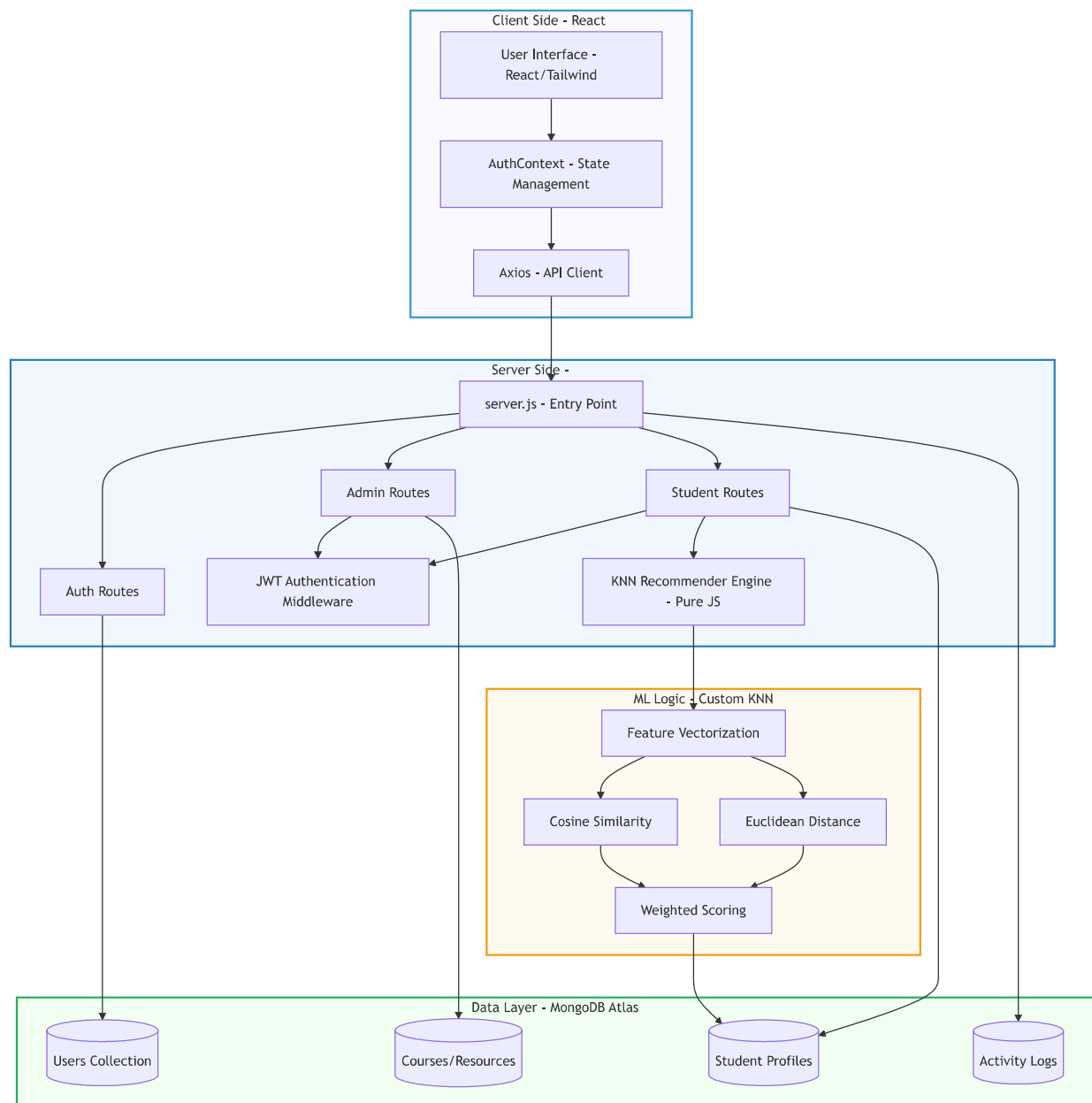
# SkillSphere System Architecture & POC

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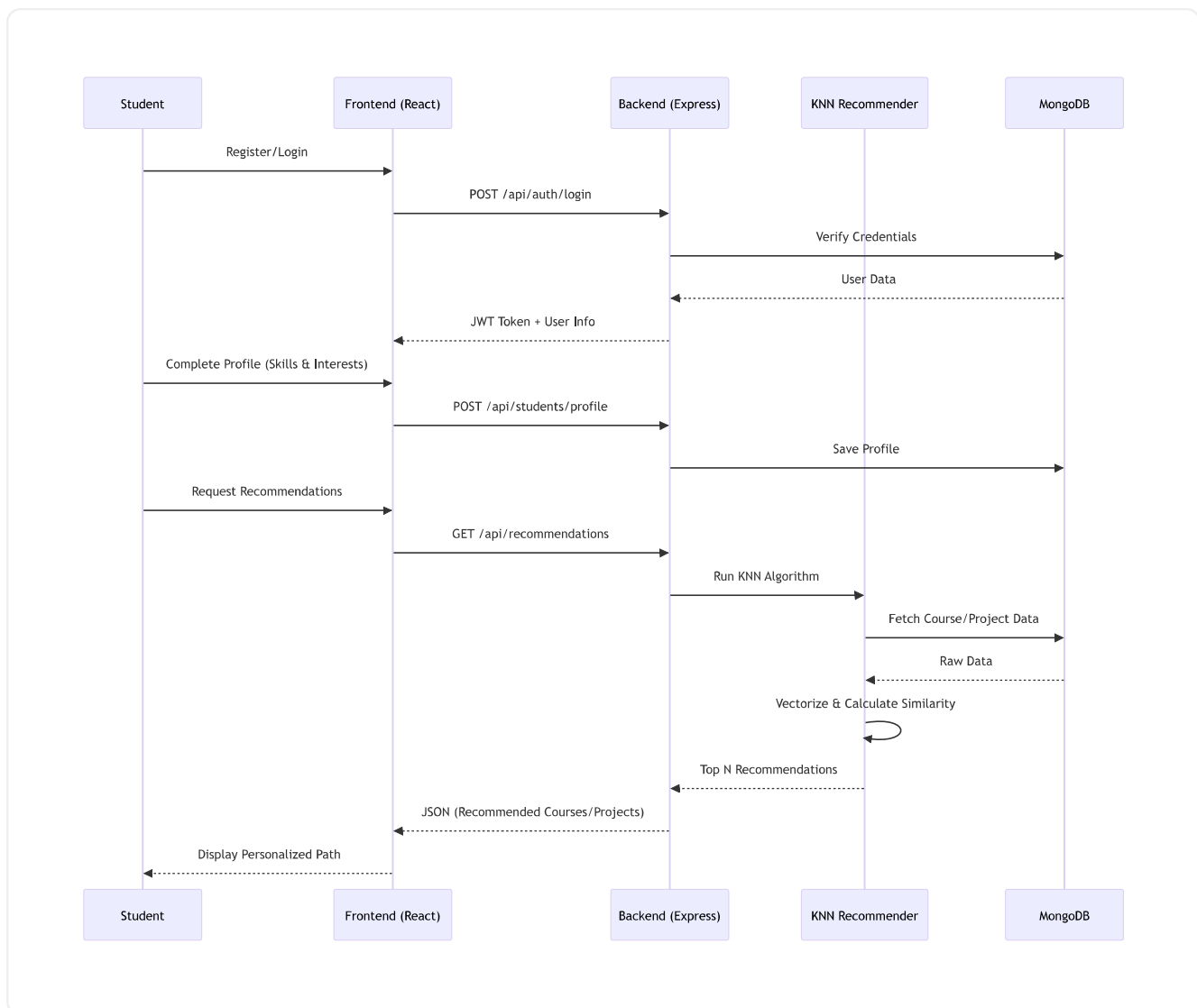
💡 **How to get JPG:** Open this file in your browser, then use "Right Click > Save Image As" (if supported) or take a high-quality screenshot (Win+Shift+S or Cmd+Shift+4) and save as JPG.



## Final System Architecture



## POC (Proof of Concept) Flow



## 🌟 POC Highlights

### Custom ML

Pure JavaScript KNN implementation with Euclidean & Cosine similarity. No external ML libraries used.

### Personalization

Analyzes student skill gaps and interest vectors to map them to the best-fit career roles.

### Admin Control

Dynamic CSV data ingestion and framework management for real-time skill mapping updates.

### Full Stack

Complete MERN stack (MongoDB, Express, React, Node) with JWT-secured authentication.



SkillSphere - Hackathon 2026 | System Architecture Documentation